SMART INDIA HACKATHON 2026

- **Problem Statement ID – SIH26058**
- **Problem Statement Title- Development of a Low-Power, Real-Time Adaptive Software-Defined Sonar Transmitter Payload for Autonomous Underwater Vehicles (AUVs)**
- **Theme- Robotics and Drones**
- **PS Category- Hardware**
- **Team ID- 138620**
- **Team Name - Hyper Grey**



HYPER GREY

SeaNergy

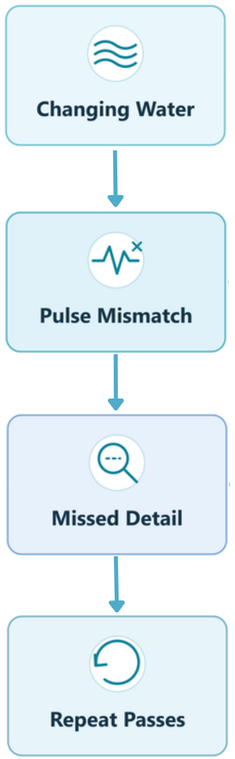
PRECISION GOES DEEPER

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demo VIDEO-Link

PROPOSED SOLUTION

SeaNergy is an **adaptive** AUV sonar payload that uses environmental data to **optimize** each ping's **waveform, frequency, bandwidth, duration, and power**. Its embedded transmitter generates the pulse, while a receiver and local console display the return signal and explain the adaptation.



Water-aware adaptation

- **Environmental sensing:** Temperature, salinity, turbidity, pH and depth describe the water around the AUV.
- **Adaptive decisions:** The controller updates frequency, bandwidth, pulse length and drive power as conditions change.

Low-power embedded transmission

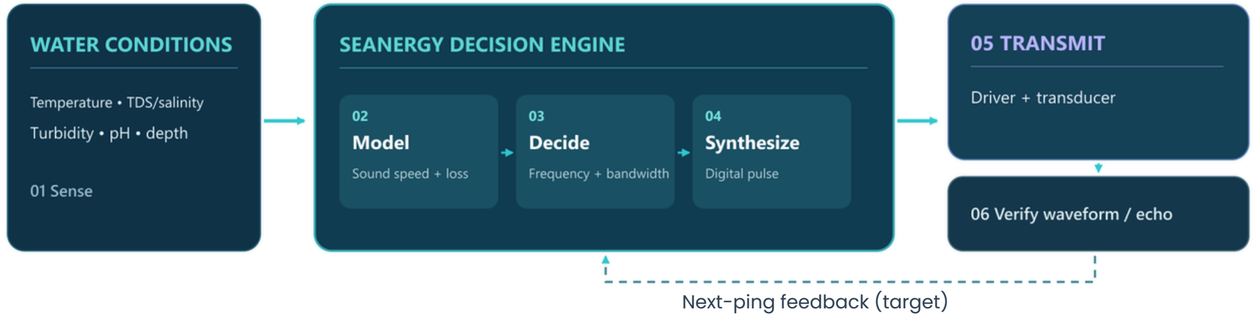
- **Bounded solver:** Acoustic, electrical and energy limits rule out impractical pulse settings before transmission.
- **Hardware output:** DDS and DMA feed the DAC, filter and driver without sample-by-sample CPU control.

Multiple waveform options

- **LFM chirps:** Frequency sweeps support pulse compression for finer range detail.
- **Geometric and phase-coded pulses:** Additional modes offer different tradeoffs in range, clutter and signal clarity.

Why the need?

Ocean survey teams use AUV sonar to map the seabed and find underwater hazards with **less reliance on survey ships**. A mission needs both **wide coverage** and **enough detail** to spot small objects, but **battery** and **survey time** are **limited**. Fixed sonar settings force a compromise as conditions change, risking missed features and repeat passes.



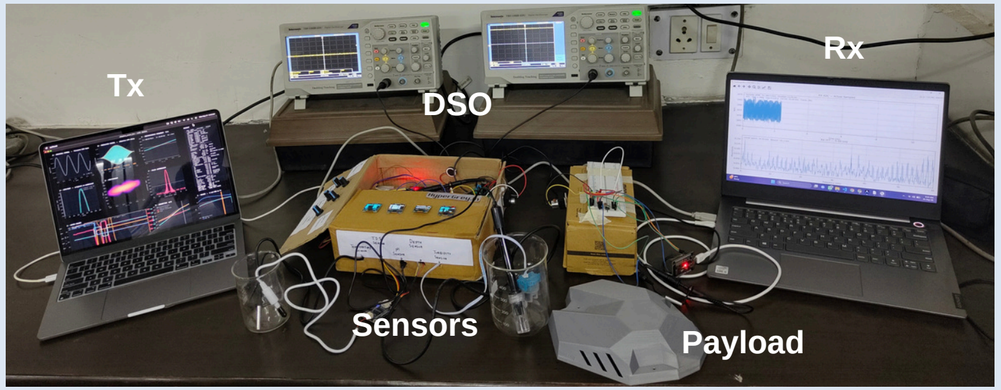
How changing water affects ranging

Water change	Approx. sound-speed shift	Illustrative error at 100 m*
+1 °C temperature	+4.5 m/s	30 cm
+1 ppt salinity	+1.3 m/s	9 cm
+100 m depth	+1.7 m/s	11 cm

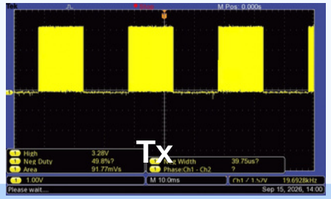
TECHNICAL APPROACH

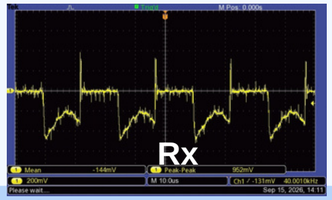
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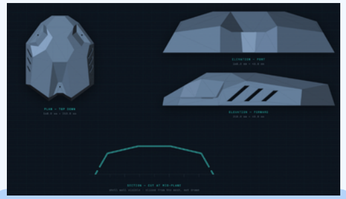


Hardware-Setup



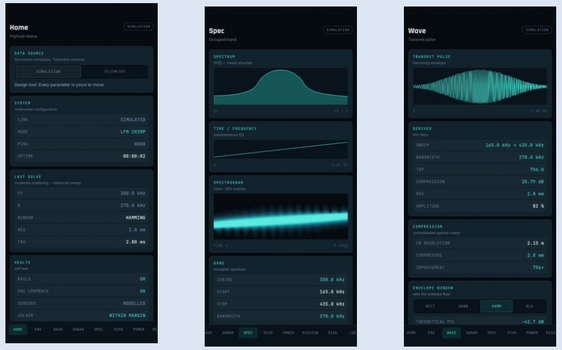


DSO- waveforms

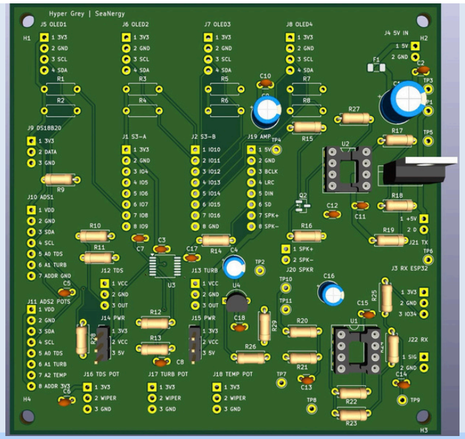


3D Model of paylaod

Click here to download app

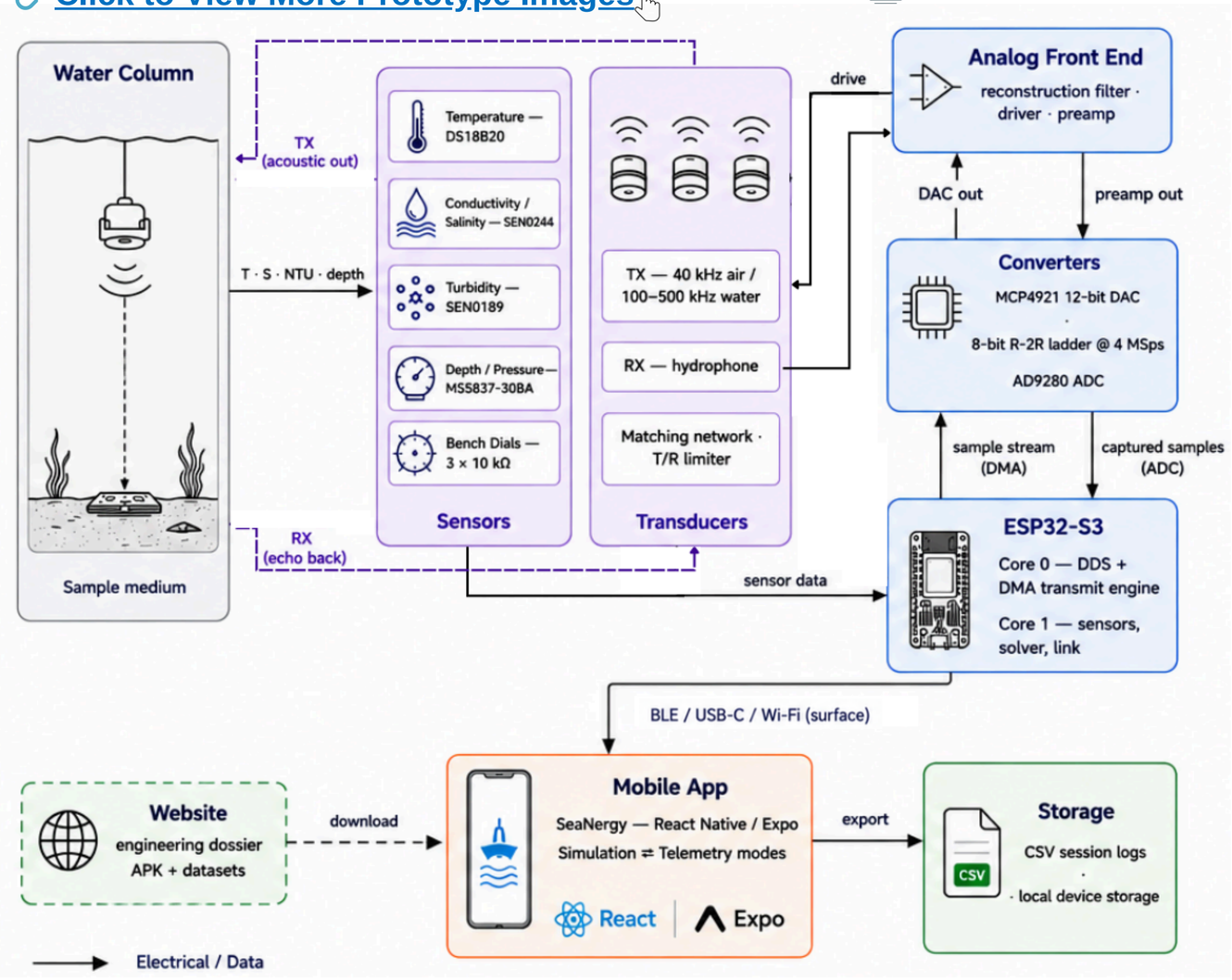


App(ios+android)



PCB

Click to View More Prototype Images



Hardware parts: Sipeed Tang Nano 9K, ESP32-S3-WROOM-1-N8R8, Sea-Bird SBE 3F, Sea-Bird SBE 4C, Seapoint STM/MCBH6M, ANB Sensors OC300, TE MS5837-30BA01, TI ADS1115IDGSR, TI DAC902, ADI AD9280ARSZ, TI OPA1652AIDR, Microchip MD0100, TI UCC27211A-Q1, TI CSD19531Q5A, TI INA226AIDGSR, ADI MAX3485ESA+, Airmar P66, Blue Robotics WetLink M10
Firmware:DMA, FreeRTOS, ESP-IDF

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FEASIBILITY AND VIABILITY

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Technical Feasibility

Open engineering stack
ESP-IDF, KiCad and auditable DSP avoid core software licence fees.

Available components
ESP32-S3, ADCs, power devices and marine transducers have supply paths.

Five-sensor autonomy
Temperature, TDS, turbidity, pH and depth guide each ping onboard.

Modular TX/RX payload
Protected receive and swappable stages allow power-level upgrades.

Operational Feasibility

Independent of cloud connectivity
The payload chooses pings onboard; operators review local logs and health data.

AUV/ROV integration pathway
Defined power, timing and data interfaces support AUV/ROV pilot installs.

Repeatable field service
Calibration, acceptance tests, serial traceability and replaceable modules.

Financial Viability

Demo planning envelope

INR 36k

INR 27,648 estimated unit cost. Shared lab equipment excluded; supplier quotes still required.

Field-payload scaling reserve

INR 14.35L

Planned power, receive, housing and calibrated measurement upgrades. Released in stages after test gates.

Risk Management & Mitigation

Potential risk	Impact	Mitigation strategy
Changing water	Range or detail may degrade	Five-sensor waveform selection; safe fallback pulse.
Power & heat	Endurance or driver damage	Current/thermal telemetry, duty caps and hardware shutdown.
TX/RX cross-talk	Missed close echoes	T/R isolation, blanking, clamp and recovery calibration.
Ingress at depth	Underwater payload loss	Pressure-rated housing, seals and immersion qualification.
Calibration drift	Unreliable echo measurements	Factory calibration, self-test and reference checks.

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Impacts & Benefits

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Key Impacts Metrics

Expected Outcomes

- ≥20% — Less energy per ping — vs fixed-power TX baseline
- ≤1 ping — Adaptation latency — after sensor / RX update
- ≥15 dB — Lower sidelobes — windowed LFM vs plain LFM
- ≥95% — Echo detection — defined tank target ≤5% false alarms
- ≤5% — Range error — known-distance tank test
- ≥99.9% — Valid TX pulses — no DMA underrun / timing fault

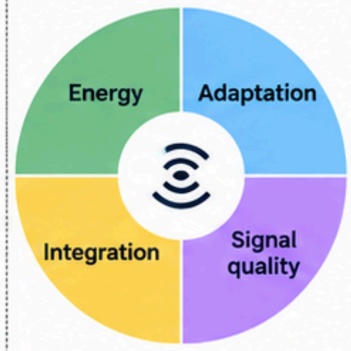
Contribution to National Goals

- Deep Ocean Mission
- NIOT marine instrumentation
- SDG 14: Life Below Water
- Blue Economy

Validation and Application



Design priorities



Multidimensional Benefits



FOR AUV OPERATORS

- Efficient sensing with lower power per mission
- Configurable mission settings (range, resolution, mode)
- Real-time system health monitoring

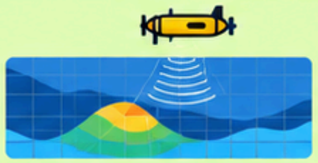
"More capable missions, longer endurance."



FOR OCEAN RESEARCHERS

- Environment-context data from adaptive sonar
- Reproducible and comparable acoustic records
- Deeper insights into marine life and habitats

"Better data for a deeper understanding."



FOR MARINE MAPPING TEAMS

- Range-resolution control for different survey needs
- Richer echo information for better seafloor maps
- Flexible operation across diverse environments

"Sharper maps, greater coverage."



FOR NATIONAL MISSIONS

- Indigenous payload development capability
- Supports underwater robotics ecosystem
- Enhanced ocean survey and monitoring capability

"Self-reliant technology for a stronger ocean future."



FOR INDUSTRY & EDUCATION

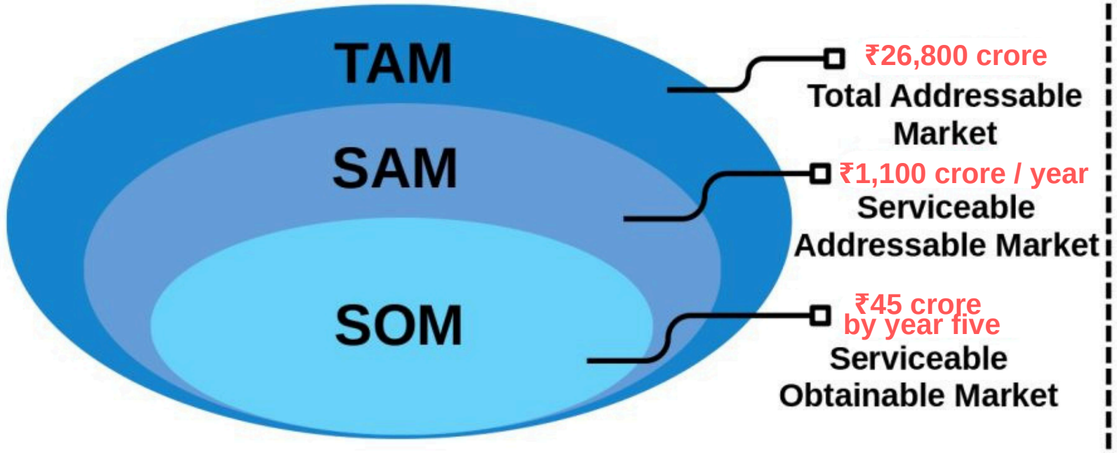
- Reusable, modular platform for varied applications
- Rapid experimentation and innovation
- Builds local expertise and skilled workforce

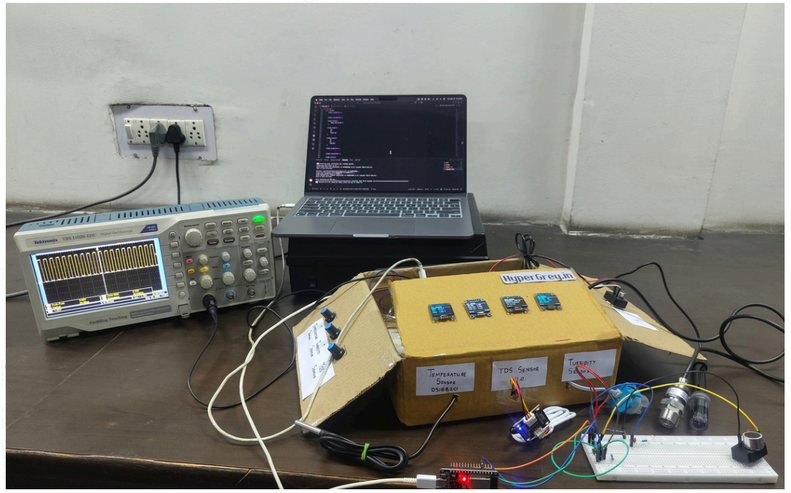
"A platform for innovation and talent."

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RESEARCH AND REFERENCES

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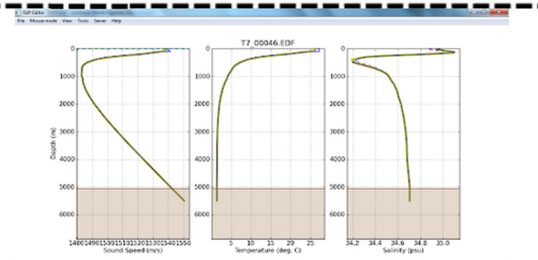


Fig: NOAA Sound Speed Temperature Profile

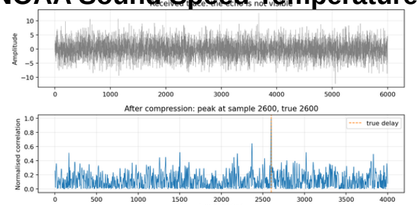


Fig: Pulse Compression Below Noise

REFERENCE

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3. High-Resolution Wideband Waveform Design for Sonar Based on Multi-Parameter Modulation — Remote Sensing 15(18), 2023 (open access): Link

4. Calculation of absorption of sound in seawater — National Physical Laboratory, UK: Link

5. A Survey of Underwater Acoustic Data Classification Methods Using Deep Learning — Sensors 22(6), MDPI (open access): Live

6. Deep Ocean Mission — Cabinet approval and technical scope — Press Information Bureau, Government of India: Live

LINKS

Research Depth

1. Documentation- Link
2. TAM,SAM,SOM Documentation-Link
3. PCB images- Link
4. Payload model images-Link
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6. Demo Video Playlist-Link

Tech

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